

Chapter 7

Integrity Constraints

(Ràng buộc toàn vẹn)



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fit@hcmus

Content

- **Overview**
- Characteristics of Integrity Constraints
- Classification
- Implementation

Overview

- **Case 1:** given a database schema:
 - GROUP (GroupID, GroupName)
 - STUDENT(StuID, StuName, GroupID)
- Constraint: A group of students must have a minimum of **two** students

Overview

- **Case 2:** Adding an attribute to GROUP:
 - GROUP (GroupID, GroupName, NumberOfStudent)
 - STUDENT(StuID, StuName, GroupID)
- What is the constraint ?

Overview

- Integrity Constraints (IC) are discovered from the semantics of the data or the representation of the data in business⁽¹⁾.
- IC is to ensure ⁽¹⁾:
 - The correctness of the data and data schema.
 - The semantic of the database schema
- When an IC is declared, all instances of a database must satisfy the IC at any time.
- An IC is discovered and declared by database designer during the design phase.
- An IC is defined on a relational schema or involves in many relational schemas.

⁽¹⁾ *Fundamental of Databases 4th, Ramez Elmasri & Shamkant B. Navathe, ISBN 0-321-12226-7, 2003*

Content

- Overview
- **Characteristics of Integrity Constraints**
 - Context
 - Content
 - Table of influence/ Integrity gate
- Classification
- Implementation

Context

- Context of an IC
 - **Relational schemas** that are likely to be violated the IC when performing data updates (adding, deleting, modifying data)

- Example (IC₁)
 - The salary of a lecturer cannot exceed his/her department head
 - Update operations that affect to the IC₁
 - Update the salary of the employee
 - Add a new employee to department
 - Appoint a new department manager
 - Context: EMPLOYEE, DEPARTMENT

Context (cont.)

- Example (IC₂)
 - The direct manager (of a lecturer) must be a lecturer in the same department
 - Update operations that affect the IC2
 - Update line manager of a lecturer
 - Add new lecturer
 - Context: EMPLOYEE

Content

- The content of an IC is presented by
 - Natural language
 - Easy to understand but lack of formal, consistency
 - Formal language
 - Concise, consistent but sometimes difficult to understand
 - Language forms
 - Relational Algebra
 - Tuples Relational Calculus
 - Pseudo code (mã giả)

Content (cont.)

- Example (IC₁)
 - Natural language
 - The salary of a lecturer cannot exceed his/her department head.
 - Formal language – using relational calculus

$$\begin{aligned} &(\forall t)(\text{GIAOVIEN}(t) \wedge (\exists s)(\text{BOMON}(s) \wedge \\ &\quad (\exists u)(\text{GIAOVIEN}(u) \wedge \\ &\quad \quad s.\text{TRUONGBM} = u.\text{MAGV} \wedge \\ &\quad \quad s.\text{MABM} = t.\text{MABM} \wedge \\ &\quad \quad t.\text{LUONG} \leq u.\text{LUONG}))) \end{aligned}$$

Content (cont.)

- Example (IC₂)
 - Natural language
 - The direct manager (of a lecturer) must be a lecturer in the same department.
 - Formal language – relational calculus

$$\begin{aligned} &(\forall t)(\text{GIAOVIEN}(t) \wedge (t.\text{GVQLCM} \neq \text{null} \Rightarrow \\ &\quad (\exists s)(\text{GIAOVIEN}(s) \wedge s.\text{MABM} = t.\text{MABM} \\ &\quad \wedge s.\text{MAGV} = t.\text{GVQLCM}))) \end{aligned}$$

Influence table/ Integrity gate

- Determine which update operations need to be checked/verified for IC when performing updates on the relations context.
- 2 types
 - Influence table for 1 IC
 - Aggregate influence table for all ICs

Influence table for 1 IC

IC title	Insert	Delete	Update
Relation 1	+	–	+ (Attribute)
Relation 2	–	+	–
...			
Relation n	–	+	–

(+) Violation of IC

(–) Not violation of IC

Aggregate influence table

	IC 1			IC 2						IC m		
	T	X	S	T	X	S	...	...	...	T	X	S
Relation 1	+	-	+	+	-	+				+	-	+
Relation 2	-	+	-									
Relation 3	-	-	+							-	+	-
...												
Relation n				-	+	-				-	-	+

Content

- Overview
- Characteristics of Integrity Constraints
- **IC Classification**
- Implementation

IC Classification

- **Group 1: IC within a single relation:**
 - Domain value constraints
 - Constraints involving multiple tuples
 - Constraints involving multiple attributes
- **Group 2: IC across multiple relations :**
 - Referential constraints (tham chiếu)
 - Multiple tuples through multiple relations (liên bộ liên quan hệ)
 - Multiple attributes through multiple relations (liên thuộc tính liên quan hệ)
 - Inference/calculated attribute/ derived attribute (thuộc tính suy diễn)
 - Relations form a cycle in relation graph (chu trình)

Group 1: IC domain constraints

- Constraints specifying values for an attribute

R	A	B	C	D
	α	α	1	1
	α	β	5	7
	β	β	12	3
	β	β	23	9

$\beta \in \{0, 1, \dots, 10\}$

- Domain value
 - Continuous (liên tục)
 - Disjointed (rời rạc)

Example 3

- Lecturer's gender must be 'Male' or 'Female'

- Context: LECTURER

- Content:

$$(\forall t)(\text{LECTURER}(t) \wedge (t.\text{GENDER} = \text{'Male'} \vee t.\text{PHAI} = \text{'Female'}))$$

or

$$\text{DOM}(\text{PHAI}) = \{\text{'Male'}, \text{'Female'}\}$$

- Influence table:

IC3	Insert	Delete	Update
LECTURER	+	-	+ (GENDER)

Example 4

- Allowance for each work in project must not exceed 20 million VND.

- Context: WORK_ON

- Content:

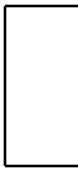
$$(\forall t)(\text{WORK_ON}(t) \wedge t.\text{ALLOWANCE} \leq 20)$$

- Influence table:

IC4	Insert	Delete	Update
WORK_ON	+	-	+ (ALLOWANCE)

Group 1: IC - Multiple tuples

- *The existence of one or more tuples depends on the existence of one or more other tuples in the same schema*

R	A	B	C	D
	α	α	1	1
	α	β	5	7
	β	β	12	3
	β	β	23	9

- Special cases
 - IC – primary key
 - IC unique (key)

Example 5

- Department name is unique.

- Context: DEPARTMENT

- Content:

$$(\forall t1, t2) (DEPARTMENT(t1) \wedge DEPARTMENT(t2) \wedge (t1 \neq t2 \Rightarrow t1.DName \neq t2.DName))$$

or

$$(\forall t1)(DEPARTMENT(t1) \wedge \neg(\exists t2)(BOMON(t2) \wedge t1 \neq t2 \wedge t1.DName = t2.DName))$$

- Influence table:

IC5	Insert	Delete	Update
DEPARTMENT	+	–	+ (DName)

Example 6

- A lecturer can participate in at most 5 works across all projects

- Context: WORK_ON

- Content:

$$(\forall t)(\text{WORK_ON}(t) \wedge \text{card}(\{s \mid \text{WORK_ON}(s) \wedge s.\text{LecturerID} = t.\text{LecturerID}\}) \leq 5)$$

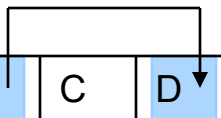
- Influence table:

IC6	Insert	Delete	Update
WORK_ON	+	−	+ (LecturerID)

Group 1: IC – Multiple attributes

- Constraint between attributes in the same relation

R	A	B	C	D
	α	α	1	1
	α	β	5	7
	β	β	12	3
	β	β	23	9



Example 8

- A lecturer must not manage him/herself (Một giáo viên không phải là người trực tiếp quản lý chuyên môn chính mình)
 - Context: LECTURER
 - Content:
$$(\forall t)(\text{LECTURER}(t) \wedge (t.\text{MgtID} = \text{null} \vee t.\text{MgtID} \neq t.\text{LecturerID})) \}$$
 - Influence table:

IC ₈	Insert	Delete	Update
LECTURER	+	–	+ (MgtID)

Example 9

- The start date of a project is always smaller than the end date.

- Context: PROJECT

- Content:

$$(\forall t)(\text{PROJECT}(t) \wedge t.\text{StartDate} \leq t.\text{EndDate})$$

- Influence table:

IC9	Insert	Delete	Update
PROJECT	+	−	+ (StartDate, EndDate)

Example 10

- The start date of a work is always smaller than the end date of that work.

- Context: WORK

- Content:

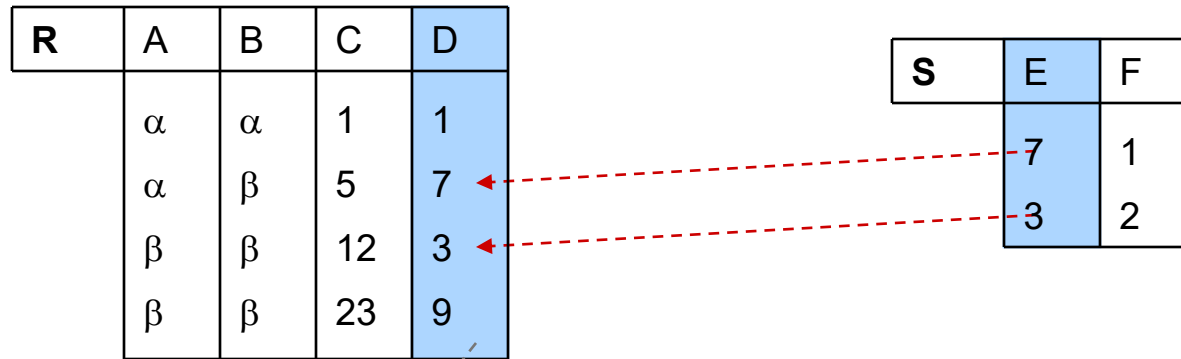
$$(\forall t)(\text{WORK}(t) \wedge t.\text{WorkSDate} \leq t.\text{WorkEDate})$$

- Influence table:

IC ₁₀	Insert	Delete	Update
WORK	+	−	+ (WorkSDate, WorkEDate)

Group 2: IC – Referential Constraints

- Values of attributes in a relation must refer to the values of attributes in other relation.



Bắt buộc phải tồn tại trước

- Special case
 - IC foreign key

Example 11

- Every lecturer must belong to exactly one department.

- Context: DEPARTMENT, LECTURER

- Content:

$(\forall t)(\text{LECTURER}(t) \wedge (\exists s)(\text{DEPARTMENT}(s) \wedge s.\text{DeptID} = t.\text{DeptID}))$

- Influence table:

IC ₁₁	Insert	Delete	Update
LECTURER	+	-	+(DeptID)
DEPARTMENT	-	+	+(DeptID)

Example 12

- Department head must be a lecturer

- Context: DEPARTMENT, LECTURER

- Content:

$(\forall t)(\text{DEPARTMENT}(t) \wedge (\exists s)(\text{LECTURER}(s) \wedge s.\text{LecturerID} = t.\text{DeptHeadID}))$

- Influence table:

IC ₁₂	Insert	Delete	Update
LECTURER	-	+	+ (LecturerID)
DEPARTMENT	+	-	+(DeptHeadID)

Group 2: IC – Referential Constraints (cont.)

- Also called existence dependency
- The context is two relations, but there are cases where it degenerates into a relation

- Example (IC₂)

- The line manager of a lecturer must be a lecturer in the same department
- Context: LECTURER
- Content:

$(\forall t)(\text{LECTURER}(t) \wedge (t.\text{MgtID} \neq \text{null} \Rightarrow (\exists s)(\text{LECTURER}(s) \wedge s.\text{DeptID} = t.\text{DeptID} \wedge s.\text{LecturerID} = t.\text{MgtID})))$

- Influence table

IC ₂	Insert	Delete	Update
LECTURER	+	+	+(MgtID, DeptID)

Group 2: IC – Multiple tuples through multiple relations

- IC is defined between tuples in multiple relations

R	A	B	C	D
α	α	1	1	
α	β	5	7	
β	β	12	3	
β	β	23	9	

S	A	B	C
α	2	7	
α	4	7	
β	2	3	
γ	2	10	

Example 13

- Each project must have at least one work

- Context: PROJECT, WORK

- Content:

$$(\forall t) (\text{PROJECT}(t) \wedge (\exists s) (\text{WORK}(s) \wedge t.\text{ProID} = s.\text{ProID}))$$

- Influence table:

IC13	Insert	Delete	Update
PROJECT	+	-	+(ProID)
WORK	-	+	+ (ProID)

Example 14

- Each department must have at least one lecturer

- Context: LECTURER, DEPARTMENT

- Content:

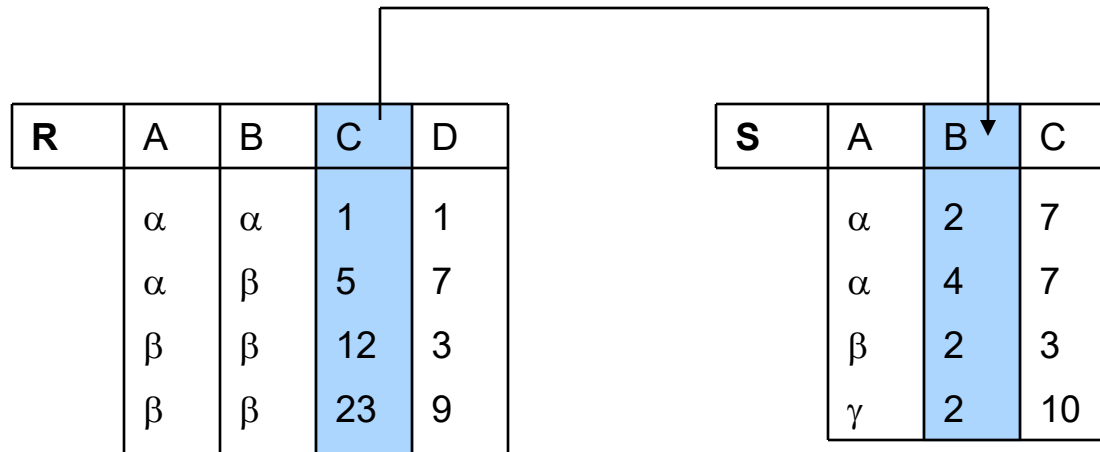
$(\forall t) (\text{DEPARTMENT}(t) \wedge (\exists s) (\text{LECTURER}(s) \wedge t.\text{DeptID} = s.\text{DeptID}))$

- Influence table:

IC14	Insert	Delete	Update
DEPARTMENT	+	-	+(DeptID)
LECTURER	-	+	+ (DeptID)

Group 2: IC – Multiple attributes through multiple relation

- IC is defined between attributes across multiple relations



Example 15

- The birthday of the department head must be less than the date of acceptance of the head position

- Context: LECTURER, DEPARTMENT

- Content:

$$(\forall t)(\text{DEPARTMENT}(t) \wedge (\forall s)(\text{LECTURER}(s) \wedge (s.\text{LecturerID} = t.\text{DeptHeadID} \Rightarrow s.\text{BirthDay} < t.\text{AppointmentDate})))$$

- Influence table:

IC ₁₅	Insert	Delete	Update
LECTURER	-	-	+ (BirthDay)
DEPARTMENT	+	-	+ (AppointmentDate, DeptHeadID)

Example 16

- Allowance of a work must be smaller than the budget of the project.

- Context: WORK_ON, PROJECT

- Content:

$$(\forall t)(\text{WORK_ON}(t) \wedge (\forall s)(\text{PROJECT}(s) \wedge (s.\text{ProID} = t.\text{ProID} \Rightarrow t.\text{Allowance} < s.\text{Budget})))$$

- Influence table:

R ₁₆	Thêm	Xóa	Sửa
WORK_ON	+	–	+ (Allowance)
PROJECT	-	–	+ (Budget)

Group 2: IC - Inference/calculated attribute

- *Calculated attribute* (also called inference attribute)
 - Its value is calculated from other attributes
- When a database has a calculated attribute
 - IC ensures the link between the calculated attributes with the source attributes is consistently maintained

Example 17

- **DEPARTMENT**(DeptID, DName, HeadDeptID, AppointmentDate, NumberOfLecturers)
- The value of the attribute NumberOfLecturers must be equal with the number of lecturers counted from departments
 - Context: LECTURER, DEPARTMENT
 - Content:

$$(\forall t)(\text{DEPARTMENT}(t) \wedge t.\text{NumberOfLecturers} = \text{card}(\{ s \mid \text{LECTURER}(s) \wedge s.\text{DeptID} = t.\text{DeptID} \}))$$

- Influence table:

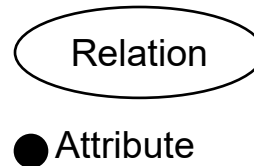
IC17	Insert	Delete	Update
LECTURER	+	+	+ (DeptID)
DEPARTMENT	+	−	+ (NumberOfLecturers)

Group 2: IC – Relations form a cycle in relation graph

■ Database schema can be presented by a graph

■ Vertex (đỉnh)

- Relation
- Attribute



■ Edge

- A line connecting a relation vertex to an attribute vertex in the database schema

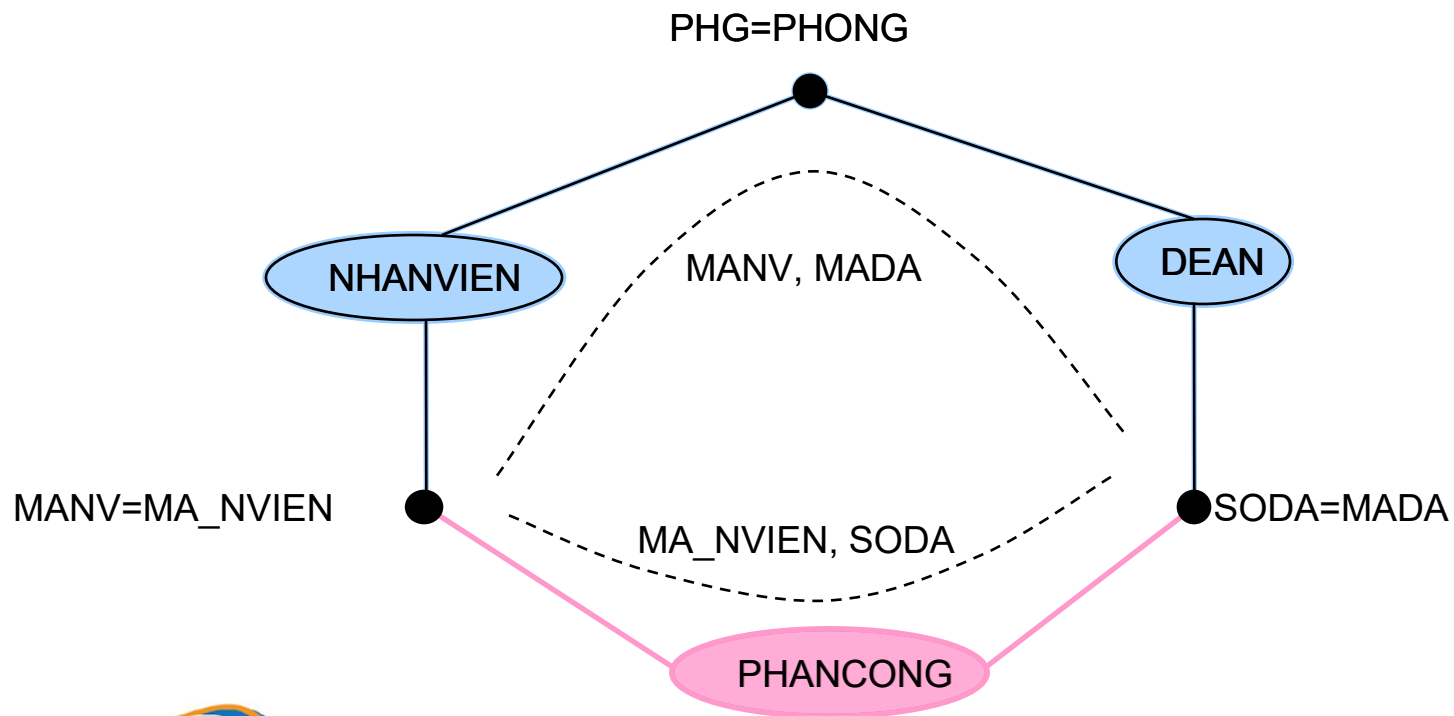


■ Cycle

- Graph appears closed path ~ Acyclic database schema

Example 17

- Employees are only assigned to projects in charged by their departments



Example 17 (cont.)

- Context: NHANVIEN, DEAN, PHANCONG
- Content:

$NVDA \leftarrow NHANVIEN \bowtie_{PHG=PHONG} DEAN$
 $(\forall t) (PHANCONG(t) \wedge (\exists s)(NVDA(s) \wedge$
 $t.MA_NVN = s.MANV \wedge t.MADA = s.SODA))$

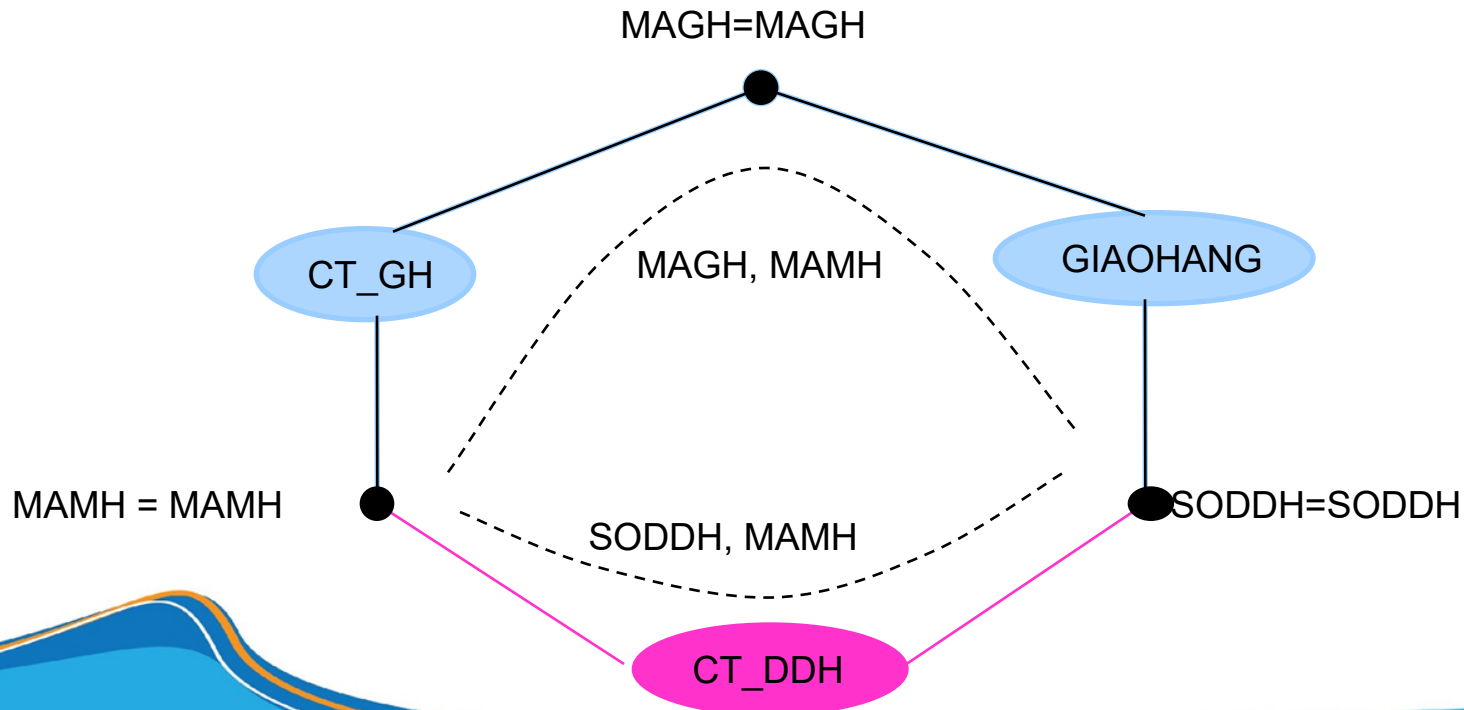
- Influence table:

IC17	Insert	Delete	Update
NHANVIEN	–	+	+ (MANV,PHG)
DEAN	–	+	+ (MADA,PHONG)
PHANCONG	+	–	+ (MA_NVN,SODA)

Example 18

- DDH (SODDH, NGAYDH, MAKH)
- CT_DDH (SODDH, MAMH, SOLUONG, DONGIA)
- GIAOHANG(MAGH, NGAYGH, TONGTIEN, SODDH)
- CT_GH (MAGH, MAMH)

IC: Only products ordered by customer are allowed to be delivered.



Content

- Overview
- Characteristics of Integrity Constraints
- Classification
- **Implementation**

Implementation

- The ICs are implemented by:
 - Primary key
 - Foreign key
 - Check constraint
 - Assertion
 - Trigger
 - Transaction
 - Stored procedure
 - Application

Assertion

- An SQL expression that always returns the value **TRUE** at all times.
 - Users need to tell what must be true
- Syntax

```
CREATE ASSERTION <Tên_assertion> CHECK (<Điều_kiện>)
```

```
DROP ASSERTION <Tên_assertion>
```

Example 15

- The department head's birthday must be earlier than the appointment date.

```
CREATE ASSERTION R12 CHECK (  
    NOT EXISTS (  
        SELECT *  
        FROM GIAOVIEN, BOMON  
        WHERE MAGV=TRUONGBM  
        AND NGAYSINH > NGAYNHANCHUC )  
    )
```

Example 19

- The head of department's salary must be greater than 50,000.

```
CREATE ASSERTION R15 CHECK (  
    NOT EXISTS (  
        SELECT *  
        FROM GIAOVIEN, BOMON  
        WHERE MAGV=TRUONGBM  
        AND LUONG < 50000 )  
    )
```

Example 19 (cont.)

- The salary of the department head must be greater than 50000

Check

Constraint

```
ALTER TABLE BOMON (  
    TENBM NVARCHAR(50) UNIQUE,  
    MABM CHAR(10) NOT NULL,  
    TRUONGBM CHAR(10),  
    NGAYNHANCHUC DATETIME,  
    CONSTRAINT CHK_BM_LUONGTRUONGBM CHECK (  
        TRUONGBM NOT IN (SELECT MAGV FROM  
GIAOVIEN  
WHERE LUONG <= 50000 ))  
)
```


Example 16

- The number of lecturers for each department should not exceed 20.

```
CREATE ASSERTION R16 CHECK (  
    20 >= ALL ( SELECT COUNT(MAGV)  
                FROM GIAOVIEN  
                GROUP BY MABM )  
)
```

Example 16 (cont.)

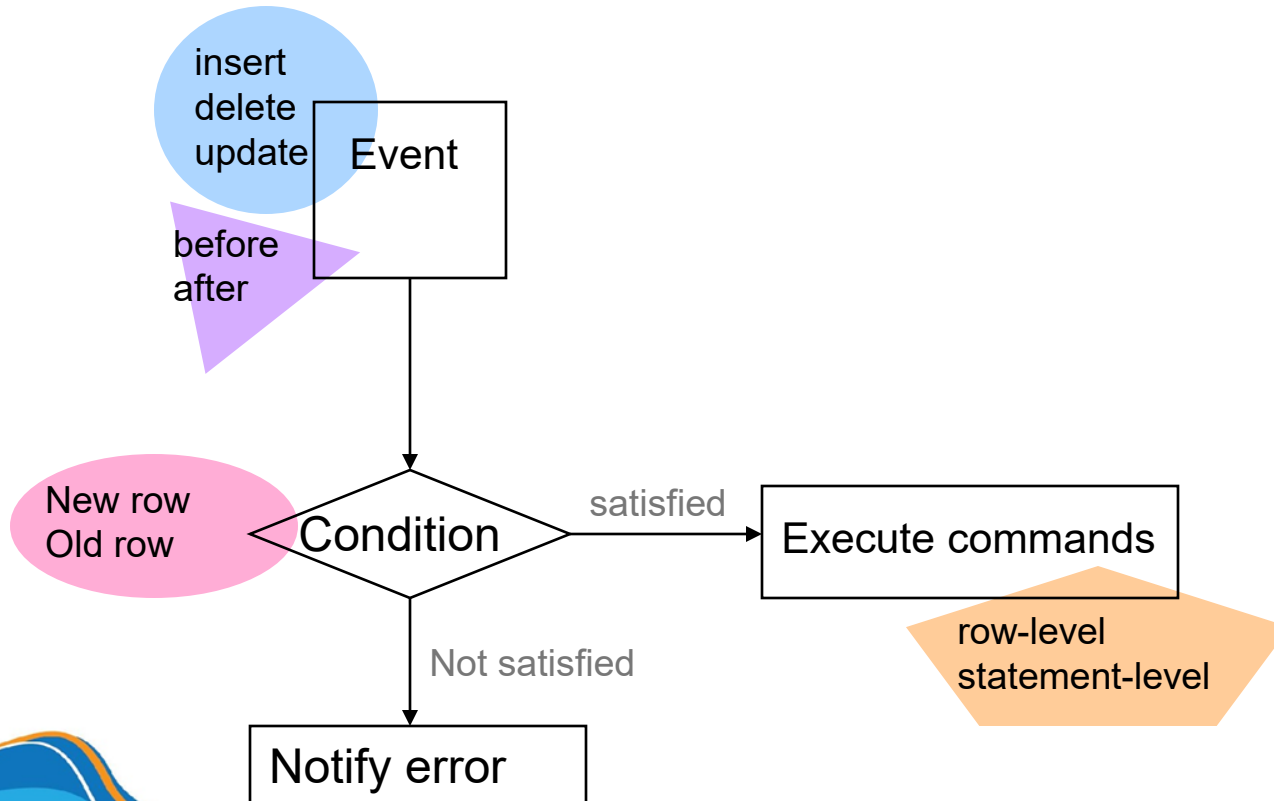
- The number of lecturers for each department should not exceed 20.

```
ALTER TABLE GIAOVIEN ADD  
CONSTRAINT CHK_GV_SLGVBM CHECK (  
    20 >= ALL ( SELECT COUNT(MAGV) FROM GIAOVIEN  
                GROUP BY MABM ))
```

Check Constraint

Trigger

- A set of commands that are automatically executed when an event occurs in the database



Trigger (cont.)

■ Syntax

```
CREATE TRIGGER <Trigger_name>  
AFTER|BEFORE INSERT|UPDATE|DELETE ON <Table>  
REFERENCING  
    NEW ROW|TABLE AS <Name_1>  
    OLD ROW|TABLE AS <Name_2>  
FOR EACH ROW | FOR EACH STATEMENT  
WHEN (<Condition>)  
    <Tập_lệnh_SQL>
```

```
DROP TRIGGER <Trigger_name>
```

Example 15

- The salary of the department head must be greater than 50000

```
CREATE TRIGGER TR_BM_UPD
AFTER UPDATE OF TRUONGBM ON BOMON
REFERENCING
    NEW ROW AS NewTuple
FOR EACH ROW
WHEN (50000 >= (SELECT LUONG          FROM GIAOVIEN
                  WHERE MAGV=NewTuple.TRUONGBM))
    Notify error to users
```

Example 15 (cont.)

- The salary of the department head must be greater than 50000

```
CREATE TRIGGER TR_BM_UPD
AFTER UPDATE OF TRUONGBM ON BOMON
REFERENCING
    NEW ROW AS NewTuple
    OLD ROW AS OldTuple
FOR EACH ROW
WHEN (50000 >= (SELECT LUONG          FROM GIAOVIEN
                  WHERE MAGV=NewTuple.TRUONGBM))
UPDATE BOMON
SET TRUONGBM=OldTuple.TRUONGBM
WHERE TRUONGBM=NewTuple.TRUONGBM
```

Example 15 (cont.)

- The salary of the department head must be greater than 50000

```
CREATE TRIGGER TR_BM_UPD
AFTER UPDATE OF LUONG ON GIAOVIEN
REFERENCING
    NEW ROW AS NewTuple
    OLD ROW AS OldTuple
FOR EACH ROW
WHEN (NewTuple.LUONG <= 50000 AND NewTuple.MAGV IN (
    SELECT TRUONGBM FROM BOMON ))
UPDATE GIAOVIEN
SET LUONG=OldTuple.LUONG
WHERE LUONG=NewTuple.LUONG
```

Transaction

- A set of instructions that perform a task/transaction in a database, such that
 - Or all instructions are executed successfully
 - Or no instruction is executed
- Ex: transfer money in a bank

Begin transaction Chuyển_tiền

Giảm tiền trong tài khoản người gửi

Tăng tiền trong tài khoản người nhận

Nếu tất cả đều thành công thì complete

Ngược lại roll back

End transaction

Transaction (cont.)

- A transaction ensure
 - Atomicity (tính nguyên tố)
 - Consistency (tính nhất quán của CSDL)
 - Related ICs are not violated
 - While executing the transaction
 - Before and after executing the transaction

Example

- Each match is a competition of exactly 2 teams

Giao tác Thêm_trận_đấu(t, s)
Thêm t vào THIDAU
Thêm s vào THIDAU
Nếu có một thao tác thất bại thì
 Quay lui giao tác
Ngược lại
 Hoàn tất giao tác
Cuối nếu
Cuối giao tác

Example

Giao tác Xóa_trận_đầu(ngay, gio)
Với mọi $s \in \text{THIDAU}$ ($s.\text{NGAY} = \text{ngay} \wedge s.\text{GIO} = \text{gio}$)
Xóa s khỏi THIDAU
Cuối với mọi
Nếu có một thao tác thất bại thì
 Quay lui giao tác
Ngược lại
 Hoàn tất giao tác
Cuối nếu
Cuối giao tác

Example

- Each invoice must have at least one invoice detailed line

Giao tác Thêm_hóa_đơn
Thêm HOADON
Thêm chi tiết thứ 1 vào CTHD
Thêm chi tiết thứ 2 vào CTHD
...
Nếu có một thao tác thêm thất bại thì
Quay lui giao tác
Ngược lại
Hoàn tất giao tác
Cuối nếu
Cuối giao tác

Example

Giao tác Thêm_hóa_đơn
Thêm HOADON
Thêm chi tiết thứ 1 vào CTHD
Thêm chi tiết thứ 2 vào CTHD
...
Nếu có một thao tác thêm thất bại thì
 Quay lui giao tác
Ngược lại
 Hoàn tất giao tác
Cuối nếu
Cuối giao tác

Stored Procedure

- DBMSs provide a way to store functions or procedures
 - Stored in the database schema
 - Used in SQL statements
- Syntax

```
CREATE PROCEDURE <Procedure_name> <para_list>  
AS  
    Local verables  
    procedure body  
GO  
EXEC < Procedure_name > < para_list values>
```

Example

- Each match is a competition of exactly 2 teams

```
CREATE PROCEDURE Thêm_trận_đấu  
t THIDAU , s THIDAU  
AS  
    begin tran  
        Thêm t vào THIDAU  
        If @@error<>0 rollback tran  
  
        Thêm s vào THIDAU  
        If @@error<>0    rollback tran  
    commit tran  
GO  
  
EXEC Thêm_trận_đấu x, y
```

Notice

- DBMS will check ICs
 - After an update operation has taken place on the database
 - At the end of each transaction
- Where should an IC be implemented?
 - DBMS
 - Application
- Too much Trigger → slow system
- Stored Procedure → high performance

